The density network: results

Evaluated as the smooth model was, on the 1,000 test spectra

P. Bhat (Fermilab), N. De Filippis (Politecnico di Bari),
K.F. Johnson, H.B. Prosper, R. Starnes (Florida State University)

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In brief

1. **Smoothness penalty.** The fine-tuned network's poorly determined functions a_{16}–a_{19} wiggled: their curvature was 10–300 times the smooth model's. A weak penalty on curvature ($\lambda = 10^{-6}$) removes the wiggles, leaves a_1–a_{15} unchanged, and slightly *improves* the accuracy. This is now the featured network.
2. **Accuracy.** It matches the smooth model above 555 GeV ($-2\ln\lambda$ per bin 1.102 against 1.100) and in the median per spectrum (1.073 against 1.078); below 555 GeV it is slightly worse (1.604 against 1.566). It is better than the old network ensemble above 555 GeV.
3. **Calibration.** With its own model-error coefficient, determined on rows not used in training, its pulls have standard deviation 1.007 and are calibrated at every degree of cancellation. Its $\varepsilon(m)$ is the smooth model's (0.07% at threshold, 0.28% at 600 GeV): two very different models need the same model-error term.
4. **The density** agrees with the smooth model's $d\sigma/dm$ to a median 0.31% (0.19% below 555 GeV), *equally at and between bin centres*; 98% of the comparisons lie within the smooth model's total $\pm 2\sigma$.
5. **Below the fitted range**, both models predict the unused 240–255 GeV bin 7% high.

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1 What was evaluated

`HEFTDensity` (SiLU, threshold factor, input $z \propto \sqrt{m - 250\ \mathrm{GeV}}$, 8 Gauss–Legendre nodes per bin). The L1 starting network `runs/DENS_L1_SiLU_thr` and seven fine-tunes from it, all with Poisson variances from the model and full-batch Adam (5,000 steps):

- four without a penalty: Huber loss of the pulls with $\delta = 1, 2, 3$, and plain χ^2 (`runs/FTD_SiLU_*_pearson`, made on the 5090);
- three with $\delta = 2$ and the smoothness penalty, $\lambda = 10^{-6}, 10^{-5}, 10^{-4}$ (`runs/FTD_SiLU_huber_d2_pearson_curv*`, made on a laptop CPU, 10 minutes each).

The featured network is $\delta = 2$, $\lambda = 10^{-6}$. The evaluation is `code/heft_density_eval.ipynb` (about 5 minutes on a laptop); the plan was `docs/density/density_eval_plan.pdf`.

2 The smoothness penalty

Bin integrals cannot see structure inside a bin, and the coefficient functions whose monomials are small or nearly degenerate over the sampled θ (a_{16}–a_{19}) are barely constrained by the data. Without a penalty the network fills that freedom with small-scale wiggles. The penalty added to the fine-tune loss is λR, with

$$R = \frac{1}{23}\sum_{i=1}^{23} R_i, \qquad R_i = \frac{\sum_j \left(g_i''(z_j)\right)^2}{\sum_j g_i(z_j)^2}, \qquad g_i = P_i\, e^{Q_i}, \tag{1}$$

the relative curvature of each function's smooth part (without the threshold factor, which is physics), in the network's input $z = 2\sqrt{(m - 250)/1205} - 1$, on 1,201 equally spaced points. This is the variable in which the spline model's smoothness prior acts (its knots are uniform in $\sqrt{m - 250}$). Dividing by $\sum g_i^2$ treats functions of very different sizes alike.

	$-2\ln\lambda$ per bin			median R_i	R_i, $i = 16 \ldots 19$	r.m.s. diff. from smooth (%)	
	255–555	> 555	median	a_1–a_{15}		a_{16}–a_{19}	a_{20}–a_{23}
smooth model	1.566	1.100	1.078	134	960, 583, 16, 426	–	–
$\lambda = 0$	1.623	1.099	1.075	141	289k, 170k, 6.2k, 4.8k	12.8	2.4
$\boldsymbol{\lambda = 10^{-6}}$	**1.604**	**1.102**	**1.073**	136	1270, 1026, 1557, 319	9.1	1.7
$\lambda = 10^{-5}$	1.612	1.102	1.076	120	315, 276, 408, 86	25.1	1.4
$\lambda = 10^{-4}$	1.665	1.104	1.090	102	56, 39, 86, 10	32.3	2.8

Table 1: The effect of the penalty ($\delta = 2$). R_i as defined above (for the smooth model, of a_i/ϕ). The r.m.s. difference from the smooth model's a_i is relative to $\max|a_i|$, on the z grid; medians over the functions in each group.

- At $\lambda = 10^{-6}$ the curvature of a_{16}–a_{19} falls by factors of 4–300, to the level of the smooth model's own, while a_1–a_{15} are untouched (median R_i 136 against 141). The accuracy improves slightly, and the network's a_{16}–a_{19} move *towards* the smooth model's.
- At $\lambda = 10^{-5}$ the network is smoother than the smooth model, and its a_{16}–a_{19} settle on a different smooth solution (25% from the smooth model's); at 10^{-4} even the well-determined functions are flattened and the accuracy suffers (1.665, 1.090).

- The differences that remain for a_{16}–a_{19} (2–8% on a 1 GeV grid for $\lambda = 10^{-6}$) are well within the smooth model's own uncertainties for these functions (median relative uncertainty 21–74%, `docs/smooth/smooth_results.pdf`).

3 Accuracy

For each bin, with n the exact POWHEG count and μ the prediction in events, the likelihood-ratio statistic $-2\ln\lambda = 2[\mu - n + n\ln(n/\mu)]$ has mean 1 for a correct model.

model	mean $-2\ln\lambda$ per bin 255–555 GeV	> 555 GeV	median per spectrum	spectra with mean > 5	predictions ≤ 0
smooth model	1.566	1.100	1.078	4	0
free fit per bin (the floor)	1.559	1.105	1.075	4	0
old network ensemble	1.561	1.121	1.085	3	3
density, L1 (start)	2.849	1.199	1.137	22	66
density, Huber $\delta = 1$	1.635	1.098	1.074	5	1
density, Huber $\delta = 2$	1.623	1.099	1.075	4	1
density, Huber $\delta = 3$	1.610	1.103	1.077	4	1
density, χ^2	1.579	1.106	1.082	5	2
density, $\delta = 2$, $\lambda = 10^{-6}$	**1.604**	**1.102**	**1.073**	4	1

Table 2: Accuracy on the 1,000 test spectra (80,000 bins).

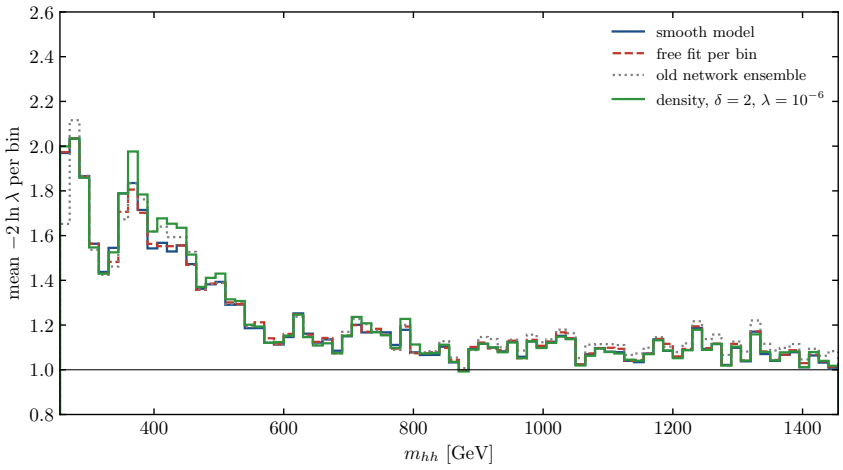


Figure 1: Mean $-2\ln\lambda$ per bin against m_{hh}.

Above 555 GeV the density network is as accurate as the smooth model and the free fit, and better than the old ensemble (trained with weights from the observed counts, and biased low there). Below 555 GeV the fine-tunes are 0.01–0.07 above the smooth model; the smaller the Huber δ, the less the fit is driven by the large pulls of the most cancelling spectra, and the larger the mean. The difference is small next to the common excess over 1 (the systematic of `docs/dihiggs_model.pdf`, section 6), and these are single seeds. The fine-tune is essential: the L1 starting network is far from the floor near threshold.

Why the fine-tune notebook gives different numbers

`heft_finetune.ipynb` reports Neyman's χ^2, $(n-\mu)^2/n$, with n estimated from N_{eff}. Using the observed count as the variance favours predictions that are biased low, and the density networks, fitted with their own Poisson variances, are not. The likelihood-ratio statistic has no such preference, which is why the paper uses it.

4 Calibration

Each model gets its own model-error coefficient $\varepsilon(m)$, determined on the 38,076 quick-test rows (never used in training) by requiring unit variance of the pulls in each bin, then smoothing $\ln\varepsilon$ by a quadratic. The network has no fit uncertainty; it is set to zero.

model, model-error term	s.d. of the pull all	255–555	> 555	$\lvert z\rvert < 2$	$\langle z^2\rangle$ by cancellation fifth (1 = strongest … 5)
smooth model, own ε	1.004	1.011	1.001	95.4%	0.97, 0.98, 1.04, 1.04, 1.00
smooth model, none	1.090	1.218	1.043	93.9%	1.70, 1.10, 1.07, 1.06, 1.01
density (featured), own ε	1.007	1.015	1.005	95.4%	1.00, 0.99, 1.04, 1.04, 1.00
density (featured), smooth model's ε	1.003	1.008	1.001	95.5%	0.98, 0.98, 1.03, 1.04, 1.00
density (featured), none	1.104	1.266	1.044	93.9%	1.83, 1.13, 1.07, 1.05, 1.01

Table 3: Pulls on the test spectra. For a standard normal, $\lvert z\rvert < 2$ holds for 95.4%. All the other fine-tunes give 1.007–1.008 with their own ε.

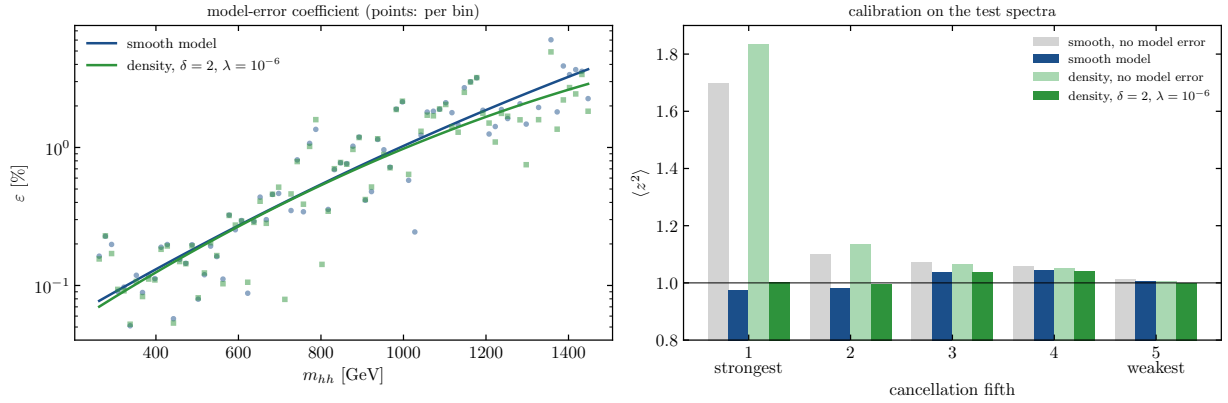


Figure 2: Left: the model-error coefficient of the smooth model and of the featured density network (points: per bin; lines: the smooth quadratic form). Right: $\langle z^2\rangle$ by fifth of cancellation, with and without the model-error term.

The two models' $\varepsilon(m)$ agree from threshold to about 1 TeV (0.070% and 0.077% at 262 GeV; 0.275% and 0.279% at 600 GeV), and the smooth model's ε calibrates the network as well as its own. Two models built in entirely different ways need the same model-error term: the systematic is a property of the POWHEG spectra, not of either model. Above 1 TeV the network's ε is lower (2.9% against 3.7% at 1.45 TeV), where both are determined from bins of about 20 events.

5 The density

Against the smooth model. On a 0.5 GeV grid from 255 to 1455 GeV, at the 1,000 test points:

run	median |network − smooth| / smooth all	at centres	between	within the smooth model's total $\pm1\sigma$	$\pm2\sigma$
density, L1 (start)	1.03%	1.03%	1.03%	57%	82%
density, $\delta = 2$, $\lambda = 0$	0.33%	0.32%	0.33%	90%	97%
density, $\delta = 2$, $\lambda = 10^{-6}$	**0.31%**	**0.30%**	**0.32%**	**90%**	**98%**

Below 555 GeV, for the featured network: 0.19%, 83% and 95%. The "total" uncertainty is the smooth model's fit uncertainty and its model-error term at the level of the density, $\varepsilon(m)\sum_i |C_i a_i(m)|$. The columns "at centres" and "between" are equal: the density network is as reliable between bin centres as at them.

Between bin centres (fig. 3). For the two spectra of moderate cancellation in the paper's `fig_interpolation`, the featured network lies within 0.6% and 2.4% of the smooth model below 600 GeV, and the four unpenalised fine-tunes differ among themselves by at most 0.16% and 0.13%, against 4.3% and 5.4% for the 20 old networks. (The four share their starting network, so their spread is a lower bound on that of a seeds ensemble.) The third spectrum passes through zero near 330 GeV, where any relative measure blows up; in absolute terms the curves agree.

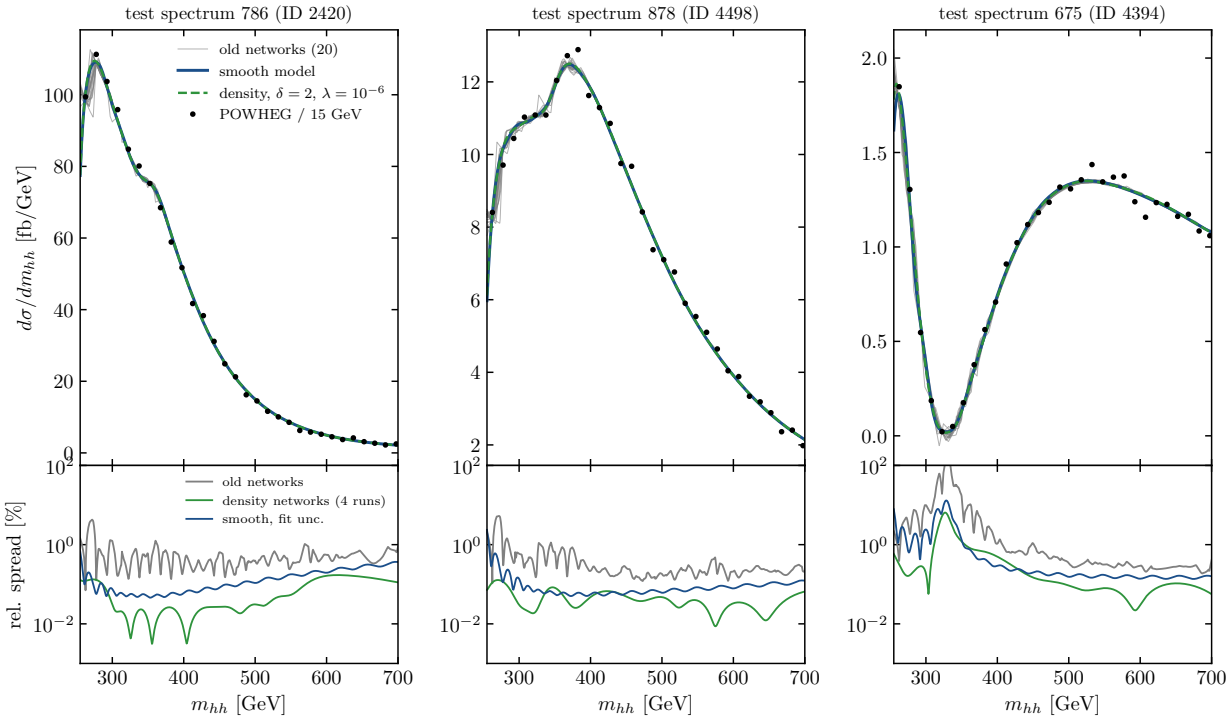


Figure 3: Three test spectra below 700 GeV: the 20 old networks (grey), the smooth model (blue) and the featured density network (green, dashed), with POWHEG divided by 15 GeV. Lower panels: relative spread of the old networks and of the four unpenalised density fine-tunes, and the smooth model's fit uncertainty.

The coefficient functions (fig. 4). On a 1 GeV grid, the r.m.s. difference between the featured network's and the smooth model's $a_i(m)$, relative to max $|a_i|$, is 0.04–0.5% for a_1–a_{15}, 1.0–1.3% for a_{20}–a_{23}, and 2–8% for a_{16}–a_{19}. The smooth model's fit bands are very narrow (a few per cent of the POWHEG noise), so even well-determined functions lie outside $\pm 2\sigma$ over parts of the range: the bands measure the fit uncertainty of one model, not the difference between two.

6 Below the fitted range

The unused POWHEG bin 240–255 GeV holds 316,626 events over the 1,000 test spectra. With the threshold factor, both models predict it as the integral of the density over 250–255 GeV:

model	observed / predicted, summed	median per spectrum
smooth model	0.937	0.957
density, L1 (start)	0.927	0.926
density, featured	0.936	0.917

Both over-predict the bin by about 7%, by the same amount: the shape just above threshold, which neither sees below 255 GeV, is extrapolated the same way.

7 Conclusions and next steps

- With a weak smoothness penalty, the density network reaches the same floor as the smooth model, needs the same model-error term, and agrees with it as a density to a few tenths of a per cent, between bin centres as well as at them, with no wiggles. It is a genuine, independent cross-check of the paper's primary model.
- Near threshold it is slightly less accurate than the smooth model (1.604 against 1.566). These are single seeds.
- **Recommended settings**: Huber $\delta = 2$, $\lambda = 10^{-6}$ (`CURV_LAMBDAS = [1e-6]` in `heft_finetune.ipynb`).
- **Next, when wanted**: a seeds ensemble of this configuration on the 5090 (fit uncertainty, spread between centres); the merged-bin test.

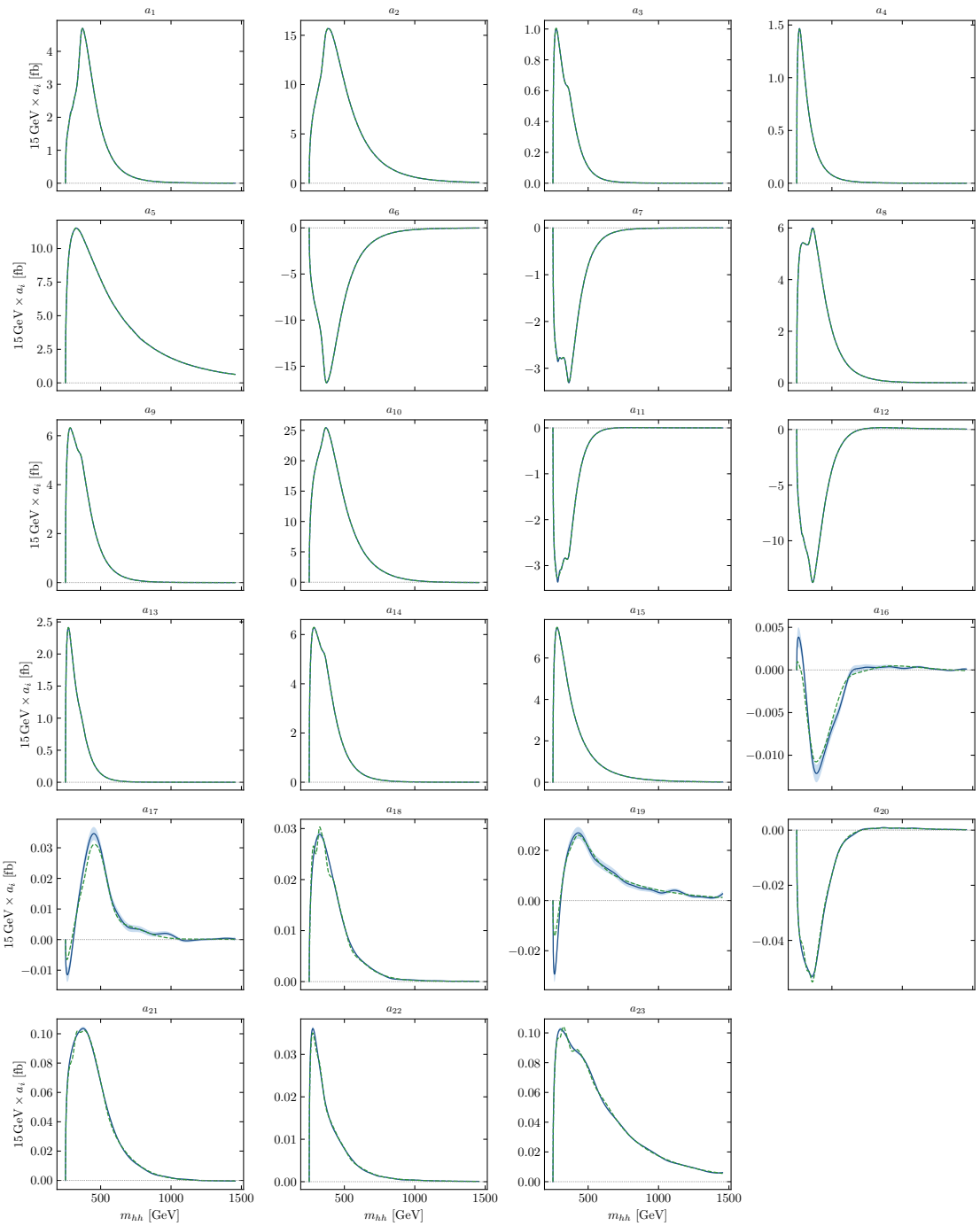


Figure 4: The 23 coefficient functions as 15 GeV $\times$ a_i: smooth model with its $\pm 1\sigma$ fit band (blue) and the featured density network (green, dashed).